Impartial Combinatorial Game Theory

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What is Impartial Combinatorial Game Theory?

- Combinatorial Games are played with perfect knowledge.
- Impartial games are games where the set of moves available, do not depend on the player currently moving, instead depending on the position. Partizan games are games which do depend on the player moving.
- We assume players will always take the best move available to them, which we will define in the following slides.

Winning and Losing

- Generally in games, the final player to make a move, wins or loses. We will look at when the final player wins.
- Intuitively, we can form an inductive hypothesis- moves that can end a game are "winning" moves, and moves that go to these moves are "losing".
- From the perspective of the current player, a winning move is a $\mathcal{N}$ position, and a losing move is a $\mathcal{P}$ position.

Example of an Impartial Game

In this game, we start from the number 7 counting down. Each player takes turns subtracting numbers 1, 2 or 3, the winner being the player to reach 0. Each player must take a turn.

Game Position	0	1	2	3	4	5	6	7
Position Type	$\mathcal{P}$	$\mathcal{N}$	$\mathcal{N}$	$\mathcal{N}$	$\mathcal{P}$	$\mathcal{N}$	$\mathcal{N}$	$\mathcal{N}$

Table: $\mathcal{P}$ and $\mathcal{N}$ positions of our game

Nim

- Nim is a fundamental game of impartial game theory, and has aided to generalisation of impartial games.
- Nim is played with piles of counters, with players taking turns removing any number from one pile— player taking the last object wins.

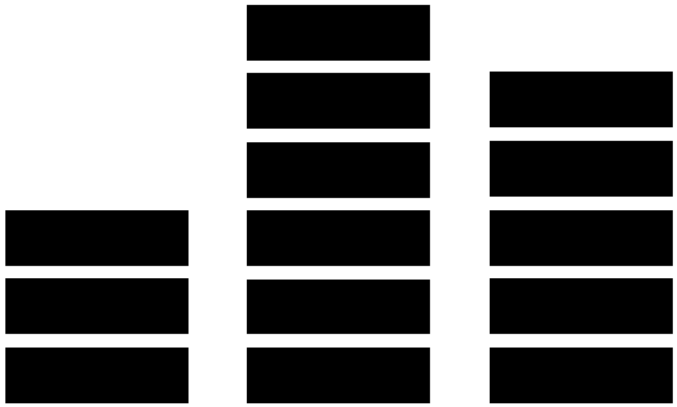


Figure: Example game of Nim using counters

Nim Proof

- Nim-sum involves the binary addition of 2 numbers, except where $1 + 1 = 0$.
- It is denoted by $\oplus$ and forms an Abelian group $(\oplus, \mathbb{N}_0)$, after converting to binary. In computer science, it is called XOR.
- We convert each of our piles to binary, and we analyse how the nim-sum of these piles change after a move is made.

	0	1	1	0
$\oplus$	1	1	0	0
	1	0	1	0

Table: XOR of 0110 and 1100

Nim Proof continued

- All nim-sum 0 positions are $\mathcal{P}$ positions, while all other positions are $\mathcal{N}$ positions.
- We prove this in two steps and induction: that Nim-sum zero positions ALWAYS leads to non-zero, and non-zero can always move to zero.

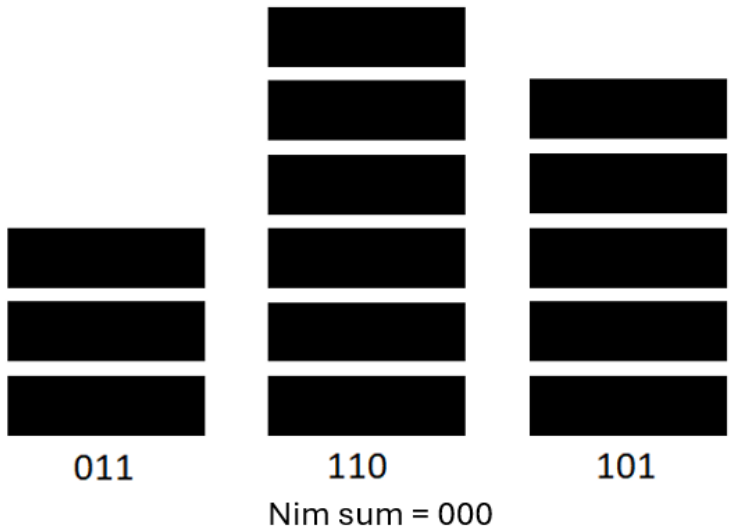


Figure: Nim-sum example

Non-Zero to Zero

- We can look at the largest offset "bit" in the Nim-sum.
- We find a pile with this bit and perform XOR with the pile and Nim-sum.
- This new value is the size of the pile after we perform our move. By doing this, we "cancel out" the bits the pile has and leave only the smaller bits (or none).

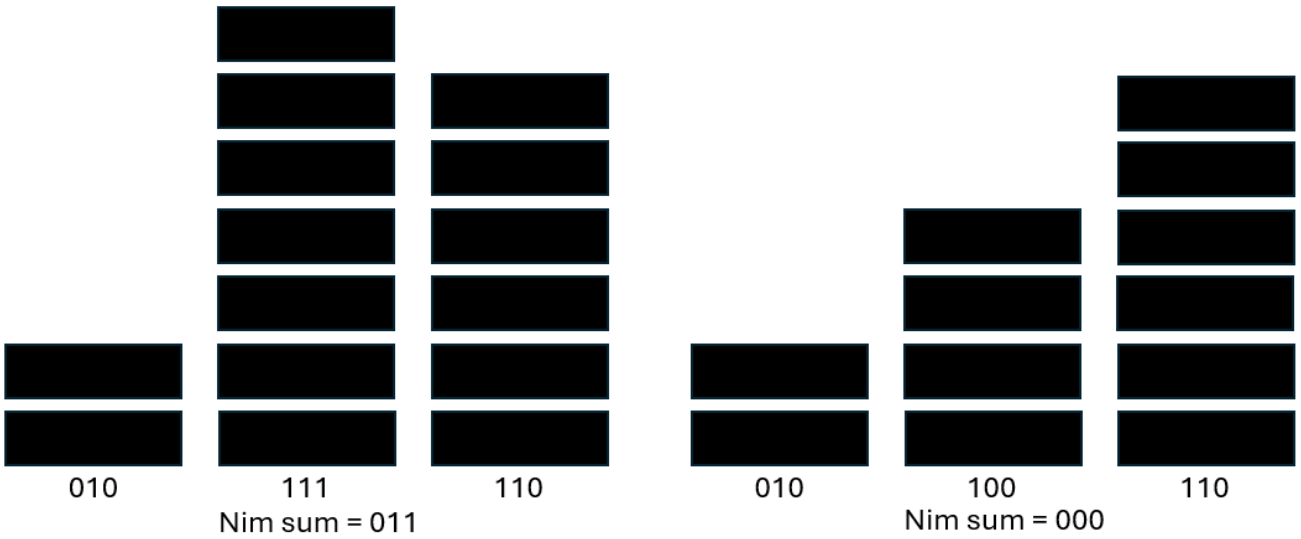


Figure: Non-zero to zero example

Zero to Non-Zero

- Conversely, nim-sum zero always leads to a non-zero nim-sum.
- Zero nim-sum means that all piles have equal "bits" which cancel each other out.
- Therefore, when a move is made and a pile is reduced, one or more bits will no longer be "cancelled out" and will appear in the nim-sum.

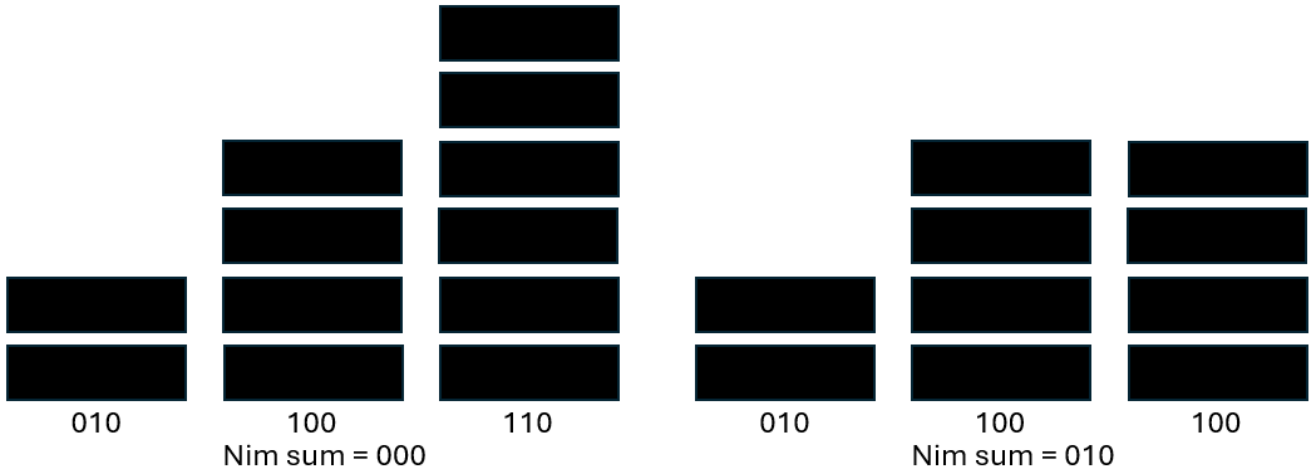


Figure: Zero to non-zero example

The Sprague-Grundy Theorem

- The Sprague-Grundy Theorem, uses nim-sum to generalise impartial games.
- The theorem states that any impartial game can be modelled as a singular Nim pile, denoted $*x$.
- Starting from the final position having nim value of $*0$, we assign nim values to the different positions of any impartial game. That is, a position that leads to $*0$ only has value $*1$, and a position that leads to $*1, *0$ has value $*2$
- This theorem then allows us to apply the proof of Nim to all other impartial games, aiding in finding $\mathcal{N}$ and $\mathcal{P}$ positions.

Thank You for Listening